Use Case Model

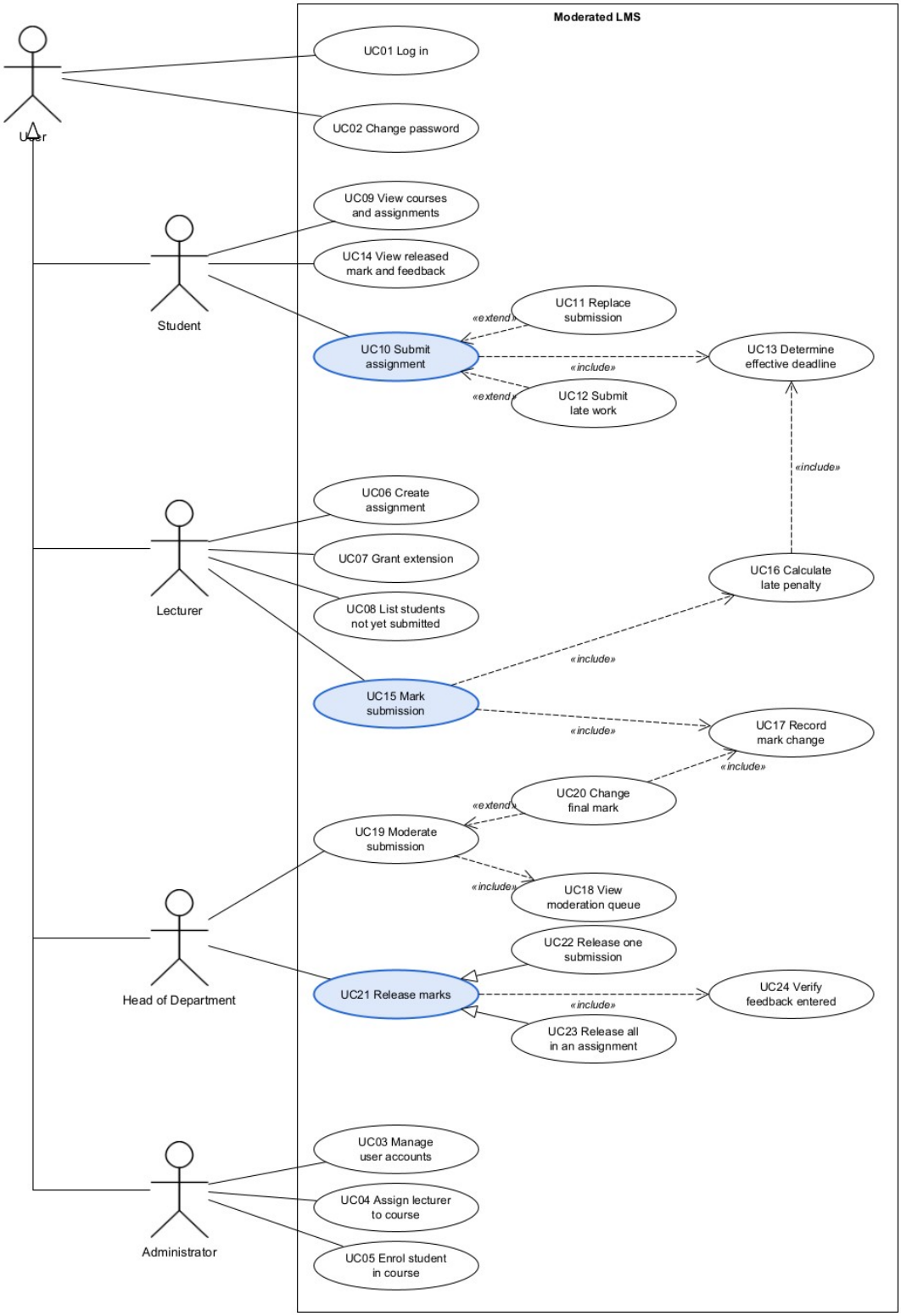
This model has 4 actors and 24 use cases, and every use case traces to a requirement in the System Requirements Specification

Actors

There are four actors, the same four roles as FR01. Each one is a kind of **User**, an abstract actor that stands for everything all of them can do (log in, change password).

Actor	Role in the system	Use cases
Student	Sees their courses, submits work, reads released marks	UC09, UC10, UC14
Lecturer	Sets assignments, grants extensions, marks work	UC06, UC07, UC08, UC15
Head of Department (HoD)	Moderates marks and decides when they are released	UC19, UC21
Administrator	Manages accounts, course assignment and enrolment	UC03, UC04, UC05
User (abstract)	Any person who can log in	UC01, UC02

Use case diagram



Read it by actor: each actor's own use cases sit in the left column, and the use cases they call or extend sit to the right. Release (UC21) is the only path to UC14, which is the Principal's rule.

Use case list

Every functional requirement FR01–FR18 is covered by at least one use case. UC10, UC15 and UC21 are specified in full below.

ID	Use case	Actor	Relationship in the diagram	Covers
UC01	Log in	User		FR01, FR02
UC02	Change password	User		FR03
UC03	Manage user accounts	Administrator		FR04
UC04	Assign lecturer to course	Administrator		FR05
UC05	Enrol student in course	Administrator		FR05
UC06	Create assignment	Lecturer		FR06
UC07	Grant extension	Lecturer		FR13
UC08	List students not yet submitted	Lecturer		FR18
UC09	View courses and assignments	Student		FR07
UC10	Submit assignment	Student	includes UC13	FR08
UC11	Replace submission	Student	extends UC10	FR10
UC12	Submit late work	Student	extends UC10	FR11, BR5
UC13	Determine effective deadline	(none)	included by UC10 and UC16	FR07, FR13
UC14	View released mark and feedback	Student		FR19, BR2
UC15	Mark submission	Lecturer	includes UC16,	FR09, FR12, FR17,

ID	Use case	Actor	Relationship in the diagram	Covers
			UC17	BR6
UC16	Calculate late penalty	(none)	included by UC15; includes UC13	FR12, BR01
UC17	Record mark change	(none)	included by UC15 and UC20	FR17, BR4
UC18	View moderation queue	(none)	included by UC19	FR14
UC19	Moderate submission	HoD	includes UC18	FR15, BR3
UC20	Change final mark	(none)	extends UC19; includes UC17	FR15
UC21	Release marks	HoD	includes UC24	FR16, BR2, BR3
UC22	Release one submission	HoD	specializes UC21	FR16
UC23	Release all in an assignment	HoD	specializes UC21	FR16
UC24	Verify feedback entered	(none)	included by UC21	FR16, BR6

UC10 Submit assignment

Field	Description
Primary actor	Student
Goal	Hand in work for an assignment so the lecturer can mark it
Interested parties	Lecturer (needs the work), Student representative (wants resubmission and late work allowed), HoD (wants the late rules applied)
Trigger	The student chooses Submit on an assignment
Preconditions	The student is logged in (UC01) and enrolled in the course. The assignment exists (UC06). The work has not been marked yet.
Postconditions (success)	One file is stored against the assignment with its

Field	Description
	submission time, and the status is Submitted (or Late). Earlier versions are kept.
Postconditions (failure)	Nothing is stored or changed, and the student sees why
Includes / extended by	Includes UC13. Extended by UC11 and UC12.
Covers	FR08, FR10, FR11, BR5
Business rules	BR5: late work is never rejected. FR10: a student may replace a submission only before the effective deadline, and the latest version counts. The effective deadline is the assignment deadline, or the later deadline if the lecturer granted this student an extension (FR13). The late penalty is applied at marking, not here (BR01).

Main success scenario (before the effective deadline, first submission)

1. The student opens an enrolled course and chooses an assignment (UC09).
2. The system works out the effective deadline (UC13) and shows it.
3. The student chooses one file.
4. The system checks that the file is 10 MB or smaller.
5. The system stores the file, records the submission time and sets the status to Submitted.
6. The system confirms the submission, showing the time and the effective deadline.

Alternative and exception flows

- **2a. The student already has a submission and the effective deadline has not passed (UC11 Replace submission).** The new file becomes the version that counts and the earlier version is kept. The student may do this any number of times until the work is marked. Continue at step 4.
- **2b. The effective deadline has passed and the student has no earlier submission (UC12 Submit late work).** The system accepts the file, never rejects it (BR5), sets the status to Late and warns that a penalty will be applied at marking. Continue at step 4.
- **2c. The effective deadline has passed and the student already has a submission.** The system refuses a second submission, because resubmission is only allowed before the deadline. The use case ends.
- **2d. The work has already been marked.** The system refuses to replace it. The use case ends.
- **4a. The file is larger than 10 MB.** The system rejects it with a message and nothing is stored. The student returns to step 3.
- **5a. The file cannot be stored (disk or database error).** The system keeps the earlier submission unchanged, tells the student the submission failed and asks them to try again. The student returns to step 3.

Notes. The penalty itself is not worked out here. It is calculated when the lecturer marks (UC16, BR01), counted from the effective deadline, so an extension from UC07 changes the result without changing this use case.

UC15 Mark submission

Field	Description
Primary actor	Lecturer
Goal	Give a submission a mark and written feedback, and send it to the HoD for moderation
Interested parties	Student (fair mark and feedback), HoD (every mark reaches the moderation queue), Principal (no mark is visible before release)
Trigger	The lecturer chooses a Submitted or Late submission to mark
Preconditions	The lecturer is logged in (UC01) and assigned to the course. The submission has status Submitted or Late.
Postconditions (success)	The awarded mark, the penalty and the final mark are stored separately. The change is logged. The status is Marked and the submission is in the HoD moderation queue. The student still sees nothing (BR2).
Postconditions (failure)	No mark, log entry or status change is saved
Includes	UC16 Calculate late penalty, UC17 Record mark change
Covers	FR09, FR12, FR17, BR1, BR2, BR4, BR6
Business rules	BR1: the penalty is 10% of the maximum mark for each started 24 hours after the effective deadline, up to 3 days, and the final mark is 0 after 3 days. BR4: every mark change is logged with who, when, old value, new value and reason. BR6: written feedback is required before a mark is released. BR2: students see nothing until the submission is Released.

Main success scenario

1. The lecturer opens one of their courses and an assignment. The system lists its submissions with their status.
2. The lecturer chooses a submission. The system shows the file, the submission time and the effective deadline.
3. The lecturer opens the file.
4. The lecturer enters the awarded mark and written feedback.
5. The system checks that the mark is between 0 and the maximum mark.
6. The system calculates the late penalty and the final mark (UC16) and shows all three values.
7. The lecturer confirms.
8. The system records the mark change (UC17): who, when, old value (none), new value and the reason "first marking".
9. The system sets the status to Marked and adds the submission to the moderation queue of the course's department.

Alternative and exception flows

- **4a. The lecturer leaves the feedback empty.** The system warns that the HoD cannot release the mark until feedback is added (UC24) and lets the lecturer save anyway or go back and write it. Feedback is preferably written before the mark is sent for moderation.
- **5a. The mark is below 0 or above the maximum mark.** The system rejects it and asks again. The lecturer returns to step 4.
- **6a. The submission was on time.** The penalty is 0 and the final mark equals the awarded mark.
- **6b. The submission is more than 3 days late.** The penalty rule gives a final mark of 0. The awarded mark is still stored and the feedback is still written (BR01).
- **9a. The submission is already Marked, Moderated or Released.** The system does not let the lecturer change the mark. After marking, only the HoD can change it (UC20).
- **9b. Saving fails.** The mark, the log entry and the status are all rolled back together, so there is never a mark without its log entry. The lecturer sees an error and tries again.

Notes. The penalty is worked out from the effective deadline, so a Grant extension (UC07) made before marking is already counted.

UC21 Release marks

Field	Description
Primary actor	Head of Department
Goal	Make marks and feedback visible to students once the department is satisfied with them
Interested parties	Principal (no mark is seen before the HoD allows it), Lecturer (students see marks soon after marking), Student (gets mark and feedback), IT officer (release must stay fast for large classes)
Trigger	The HoD chooses to release marks for an

Field	Description
	assignment in their department
Preconditions	The HoD is logged in (UC01). The submissions are in the HoD's department and have status Marked or Moderated.
Postconditions (success)	Each released submission has status Released, and its approval (who, when) is recorded. Students can now see the mark and feedback (UC14).
Postconditions (failure)	A submission that cannot be released keeps its old status, and the HoD is told which one and why
Includes	UC24 Verify feedback entered
Specialised by	UC22 Release one submission, UC23 Release all in an assignment
Covers	FR16, BR2, BR3, BR6
Business rules	BR2: students see a mark and its feedback only when the submission is Released. BR3: a submission must be Moderated (approved or changed) before it is Released. BR6: written feedback is required before a mark is released.

Main success scenario

1. The HoD opens the assignments of their department that have Marked or Moderated submissions.
2. The HoD chooses one submission to release (UC22) or all of an assignment's eligible submissions in one action (UC23).
3. The system verifies that each chosen submission has written feedback (UC24).
4. For each Marked submission, the system records the HoD's approval (who, when) in the same audit log as mark changes (FR17) and sets it to Moderated, because releasing a Marked submission counts as approving it (BR3). The HoD does not have to open every mark.
5. The system sets each chosen submission to Released.
6. The system tells the HoD how many submissions were released.

Alternative and exception flows

- **1a. No submission is waiting.** The system says so and the use case ends.
- **2a. The HoD first wants to check or change a mark.** The HoD uses UC19 Moderate submission, then returns to step 2.

- **3a. A submission has no feedback (UC22).** The system refuses to release it and says feedback is missing. The submission stays as it was. The use case ends.
- **3b. Some submissions have no feedback (UC23).** The system releases the ones that have feedback and shows the HoD a list of those skipped, so the lecturers can add feedback and the HoD can release them later.
- **4a. A submission changed status since the HoD opened the list** (for example the lecturer corrected the mark). The system skips it and lists it, and the HoD can open it again.
- **5a. Saving fails for a submission.** That submission's approval and status change are rolled back together. The others are unaffected and the failed one is listed.

Notes. Release is the only step that makes a mark visible to a student. This is the Principal's rule, settled in conflict 1 of the stakeholder analysis: the lecturer marks first, the work waits in the HoD's queue, and the HoD holds the authority to release without having to review every mark.